Predictive Delinquency Model Plan

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1. Model Logic

The predictive model is designed to identify customers who are at risk of delinquency based on historical financial and behavioral data. A Logistic Regression model was selected because delinquency prediction is a binary classification problem where the outcome is either delinquent or non-delinquent.

The model would use customer variables such as:

- Missed payments
- Credit utilization
- Credit score
- Debt-to-income ratio
- Employment status
- Income level

Predictive Workflow

- Collect customer financial data.
- Clean missing and inconsistent values.
- Convert categorical variables into numerical format.
- Split data into training and testing datasets.
- Train the Logistic Regression model.
- Generate delinquency probability scores.
- Classify high-risk customers based on prediction thresholds.

Top 5 Input Features

- Missed Payments
- Credit Utilization
- Credit Score
- Debt-to-Income Ratio
- Income Level

2. Justification for Model Choice

Logistic Regression was selected because it is simple, interpretable, and widely used in financial risk prediction. The model is effective for binary classification problems such as predicting whether a customer will become delinquent.

The model provides transparency by allowing analysts to clearly understand how each variable affects delinquency risk. This is important in financial services where explainability and regulatory compliance are critical.

Compared to more complex models such as neural networks, Logistic Regression is easier to implement, monitor, and explain to business stakeholders. It is also suitable for Geldium's requirement of identifying key risk indicators while maintaining operational simplicity.

3. Evaluation Strategy

The model would be evaluated using multiple performance metrics including accuracy, precision, recall, F1 score, and AUC-ROC.

- Accuracy measures overall prediction correctness.
- Precision evaluates how many predicted delinquent customers were actually delinquent.
- Recall measures how effectively the model identifies high-risk customers.
- F1 score balances precision and recall.
- AUC-ROC evaluates the model's ability to distinguish between delinquent and non-delinquent customers.

Bias detection would also be performed to ensure the model does not unfairly impact specific customer groups based on age, employment status, or income level.

Ethical considerations include maintaining customer privacy, avoiding discriminatory predictions, and ensuring predictions are used responsibly during lending decisions.